

Eduardo A. Pereyra

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EDUCATION

Rice University | *Houston, TX*

GPA: 3.90

Bachelor of Science in Mechanical Engineering – Trustee Distinguished Scholar

Expected Graduation Date: May 2028

Relevant Coursework: Computational Fluid Mechanics, Thermal Systems Design, Heat Transfer, Dynamics, Stress Analysis

PROFESSIONAL EXPERIENCE

Microfluidics Research Assistant | Rice University | *Houston, TX*

February 2026 – Present

- Designed and fabricated microfluidic chips, evaluating designs via manual fluid mechanics and heat transfer calculations.
- Built an Arduino PID feedback loop utilizing AC electrothermal heating to mix and maintain a 14-microliter droplet within an error of 0.25 Celsius over 30 minutes.
- Conducted CFD and thermal flow simulations utilizing ANSYS Fluent to optimize system design and validate results.
- Built MATLAB-based computer vision algorithm to automate cell counting, significantly improving accuracy and processing time.

Midstream Engineering Intern | Phillips 66 | *Houston, TX*

May 2026 – August 2026

- Developed a data-driven hydraulic model calibration workflow for gas gathering systems to reduce manual tuning and improve pressure drop predictions.
- Built an Excel-based tool utilizing SCADA data to identify optimal operating days for model calibration across three regions.
- Analyzed field data to derive a velocity-based correlation for pipe efficiency, improving model accuracy.
- Evaluated pipe equations and established a standardized roughness value to minimize average pressure error across models.

Petroleum Engineering Intern | Chevron | *Houston, TX*

May 2025 – August 2025

- Worked with the Eastern Mediterranean Business Unit to evaluate development alternatives for the Dalit offshore gas field, assessing economic viability to recommend an optimized development plan for the 2030s.
- Optimized production by refining reservoir simulation models, conducting sensitivity analyses, and integrating surface network.

Petroleum Engineering Research Assistant | The University of Tulsa | *Tulsa, OK*

May 2022 – September 2023

- Conducted a literature review on rotary gas separator geometry, presented findings to the sponsor company, and published a paper.
- Designed modifiable impeller and inducer assemblies in Autodesk Fusion 360 and 3D printed them via FDM and SLA methods.
- Designed and programmed ultra-affordable Arduino-based methane tracking system to track environmental methane emissions.

PROJECTS & LEADERSHIP

Internal Vice President | Rice Society of Hispanic Professionals and Engineers (SHPE)

September 2024 – Present

- Manage internal affairs and operations for 50+ active members within a nationally recognized Silver Chapter.
- Lead bi-weekly General Body Meeting presentations and design engaging meeting activities to foster community building.
- Collaborate with the External Vice President to integrate corporate sponsors and facilitate professional networking.

External Vice President | Rice American Society of Mechanical Engineers (ASME)

April 2025 – Present

- Created the chapter's inaugural sponsorship packet, driving corporate outreach for on-campus recruiting.
- Coordinate diverse professional and social programming, including panels, resume review workshops, and networking sessions.

Rice Recreation Center Busyness Monitor | Rice University

January 2025 – May 2026

- Built an occupancy monitoring device with a Raspberry-Pi, thermal camera, and OpenCV, deploying data to a website.
- Designed a 3D-printed SolidWorks housing with a custom cooling system to maintain temperatures below 100 degrees Fahrenheit, and assisted in developing counting algorithm with in 10% error over 10 minutes.

PUBLICATIONS

Botero, N., Pereyra, E. A., Pereyra, E., Ratkovich, N., Sarica, C. | "Review of influence of geometry on rotary gas separators." | *Taylor and Francis, Separation Science and Technology* | March 18, 2025

SKILLS / ACHIEVEMENTS

Technology: 3D CAD (Autodesk Fusion 360, SOLIDWORKS) | ANSYS Fluent | 3D Printing (FDM, SLA) | Arduino, Raspberry Pi | C | C++ | Java | JavaScript | Python | MATLAB | LaTeX | AutoCAD | Excel | Petrel | SLB Intersect | GAP | Pi Vision | SCADA | NextGen

Certifications: Certified SOLIDWORKS Associate (CSWA) | Autodesk Certified User: Fusion 360

Spoken Languages: Verbal and written fluency in English and Spanish

Achievements: 1st Place Microsoft-Sponsored Texas A&M SHPEathon Engineering Design Competition (2026)